

LABORATORY EXERCISE

1. Do the Python implementation up to nanosecond

$$\left(1 + \frac{1}{n}\right)^n$$

How often	n	$\left(1 + \frac{1}{n}\right)^n$
yearly	1	2.000000
twice a year	2	2.250000
quarterly	4	2.441406

Draw a Histogram in Matplotlib.

2. Create the Python implementation of the table:

Here is

$$\frac{a^h - 1}{h}$$

for three different bases as h shrinks.

h	$a = 2$	$a = 2.71828 \dots$	$a = 3$
0.1	0.7177	1.0517	1.1612
0.01	0.6956	1.0050	1.1047
0.001	0.6934	1.0005	1.0992
0.0001	0.6932	1.00005	1.0987
settles at	0.6931	1.0000	1.0986

Tolerance $\times 10^{-6}$

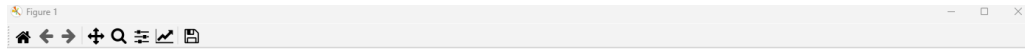
Draw a Histogram in Matplotlib.

3. Do the Python implementation of the following:

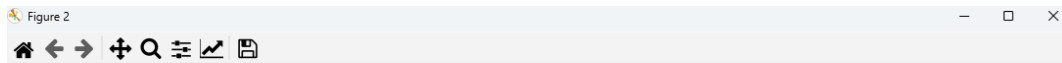
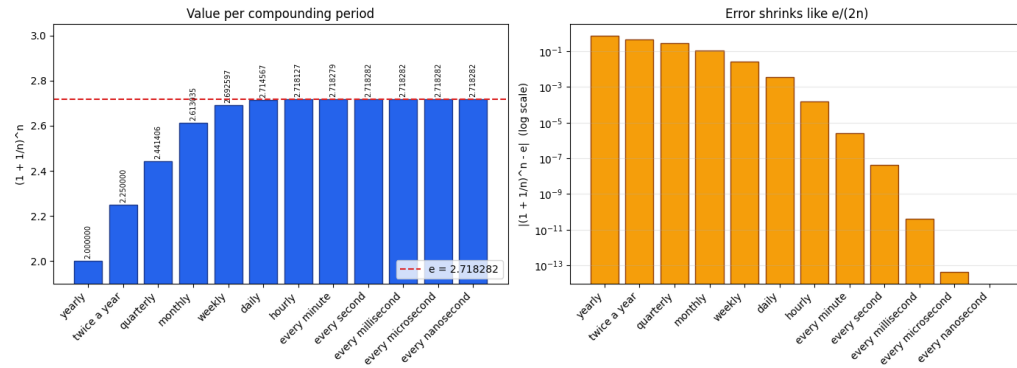
$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

Up to 10,000

Draw a Histogram in Matplotlib.

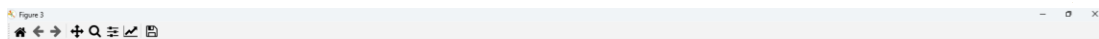
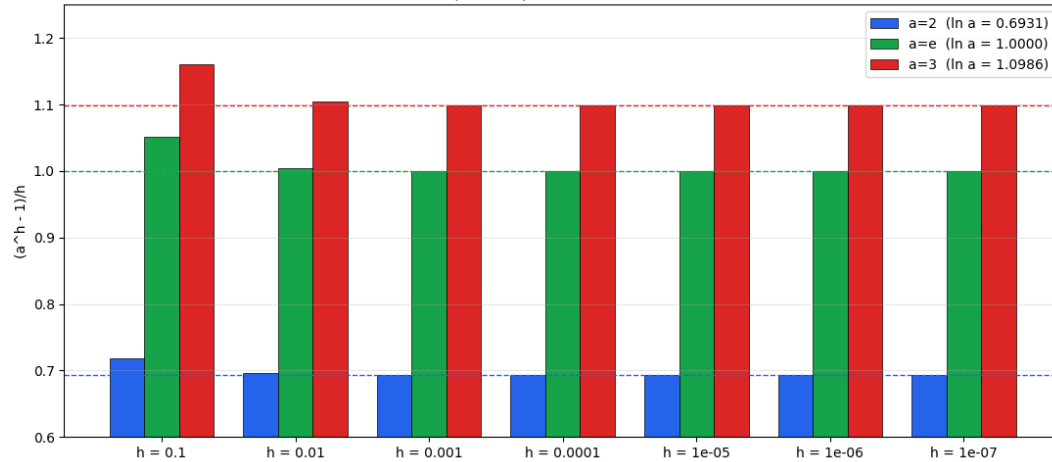


Exercise 1: Convergence of $(1 + 1/n)^n$ to e



Exercise 2: $(a^h - 1)/h$ settling to $\ln(a)$

Bars: difference quotient per h . Dashed lines: the limit $\ln(a)$.



Exercise 3: $e^x = \sum_{n=0}^{\infty} x^n / n!$ ($x = 1$), summation up to $N = 10,000$ terms

